Problem 1

A particle is confined in a one-dimensional infinite potential well of width L, with infinite potential outside the region $0 < x < L$. The normalized stationary states and energy levels are given by:

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \ldots$$

At time $t = 0$, the wavefunction of the particle is given by:

$$\psi(x, 0) = \mathcal{N} \cdot \left(\sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{3\pi x}{L}\right) + 2\sin\left(\frac{5\pi x}{L}\right)\right)$$

where $\mathcal{N}$ is a normalization constant.

1. $(\sqrt{\frac{L}{2}}N)^2 \times 2 + (2\sqrt{\frac{L}{2}}N)^2 = 3LN^2 = 1$, $N = \sqrt{\frac{1}{3L}}$
3. $\psi(x,t) = \frac{1}{\sqrt{6}}\phi_1 e^{-iE_1t/\hbar} + \frac{1}{\sqrt{6}}\phi_3 e^{-iE_3t/\hbar} + \sqrt{\frac{2}{3}}\phi_5 e^{-iE_5t/\hbar}$
4. $P_1 = |\frac{1}{\sqrt{6}}| = \frac{1}{6} = P_3$. $P_5 = |\sqrt{\frac{2}{3}}|^2 = \frac{2}{3}$
5. $|\psi(x,t)|^2 = \sim$ dependent
6. $\langle \hat{H} \rangle = \frac{1}{6}E_1 + \frac{1}{6}E_3 + \frac{2}{3}E_5$

Tasks:

1. Normalize the wavefunction and determine the value of $\mathcal{N}$.
2. Express $\psi(x, 0)$ in terms of the stationary states $\phi_n(x)$.
3. Write the time-dependent wavefunction $\psi(x, t)$.
4. Compute the probability of measuring each of the energies E_1, E_3, and E_5.
5. Determine whether $|\psi(x, t)|^2$ is time-dependent or not. Explain briefly.
6. Compute the expectation value of the energy $\langle \hat{H} \rangle$ in this state.

Problem 2

A quantum system has a time-independent Hamiltonian with the following eigenstates and eigenvalues:

$$\hat{H}\,|+\rangle = E_+\,|+\rangle\,, \quad \hat{H}\,|-\rangle = E_-\,|-\rangle$$

with eigenvalues:

$$E_+ = \hbar\omega, \quad E_- = 2\hbar\omega$$

The initial state at time $t = 0$ is given by:

$$|\psi(0)\rangle = (1 - i)\,|+\rangle + (2 + i)\,|-\rangle$$

Tasks:

1. Normalize the initial state $|\psi(0)\rangle$.
2. Write down the time-evolved state $|\psi(t)\rangle$.

1. $\sqrt{|1-i|^2 + |2+i|^2} = \sqrt{7}$

   $|\psi(0)\rangle = \frac{1}{\sqrt{7}}((1-i)|+\rangle + (2+i)|-\rangle)$
2. $|\psi(t)\rangle = \frac{1-i}{\sqrt{7}} e^{-i\omega t}|+\rangle + \frac{2+i}{\sqrt{7}} e^{-2i\omega t}|-\rangle$

Problem 3

Consider a two-qubit system in the following entangled state:

$$|\psi\rangle_{AB} = \frac{1}{\sqrt{10}}\left(2\,|00\rangle + i\,|01\rangle + |10\rangle\right)$$

Tasks:

1. Normalize the state $|\psi\rangle_{AB}$ if necessary.
2. Write down the density matrix $\rho_{AB} = |\psi\rangle_{AB}\langle\psi|_{AB}$ in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.
3. Compute the reduced density matrix $\rho_A = \mathrm{Tr}_B(\rho_{AB})$, i.e., trace out Bob's qubit.
4. Determine whether ρ_A is a pure or mixed state. Justify your answer.

$\mathrm{Tr}_B(|A_1B_1C_1\rangle\langle A_2B_2C_2|) = |A_1C_1\rangle\langle A_2C_2| \otimes \langle B_2|B_1\rangle$

1. $|\psi_{AB}\rangle = \frac{1}{\sqrt{6}}(2|00\rangle + i|01\rangle + |10\rangle)$

2. $|\psi_{AB}\rangle\langle\psi_{AB}| = \frac{1}{6}\begin{pmatrix}2\\ i\\ 1\\ 0\end{pmatrix}(2, -i, 1, 0)$

$\begin{pmatrix}1&0\\0&0\end{pmatrix}$ $|0\rangle\langle 0| \otimes \langle 0|0\rangle$

$= \frac{1}{6}\Big(4|00\rangle_{AB}\langle 00| - 2i|00\rangle\langle 01| + 2|00\rangle\langle 10| + 2i|01\rangle\langle 00| + |01\rangle\langle 01| + i|01\rangle\langle 10| + 2|10\rangle\langle 00| - i|10\rangle\langle 01| + |10\rangle\langle 10|\Big)$

$= \frac{1}{6}\begin{pmatrix}4 & -2i & 2 & 0\\ 2i & 1 & i & 0\\ 2 & -i & 1 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$

3. $\mathrm{Tr}_B(\rho_{AB}) = \frac{1}{6}\big(4|0\rangle\langle 0| + 2|0\rangle\langle 1| + |0\rangle\langle 0| + 2|1\rangle\langle 0| + |1\rangle\langle 1|\big)$

$= \frac{1}{6}\begin{pmatrix}5 & 2\\ 2 & 1\end{pmatrix}$

4. $\rho_A^2 = \sim \neq \rho_A$

$\mathrm{Tr}(\rho_A^2) =$ 对角线之和 < 1 mixed

Problem 4

Consider the following normalized three-qubit state:

$$|\psi\rangle_{ABC} = \frac{1}{\sqrt{3}}\left(|001\rangle + |010\rangle + |100\rangle\right)$$

Tasks:

1. Compute the reduced density matrix $\rho_{AC} = \mathrm{Tr}_B(\rho_{ABC})$, i.e., trace out the middle qubit B.
2. Determine whether the reduced state ρ_{AC} is mixed or not.

1. $|\psi\rangle_{ABC}\langle\psi|_{ABC} = \frac{1}{3}\big(|001\rangle\langle 001| + |001\rangle\langle 010| + |001\rangle\langle 100| + \cdots\big)$

$\rho_{AC} = \mathrm{Tr}_B(\rho_{ABC}) = \sim$

2. $\rho_{AC}^2 \sim \rho_{AC}$, $\mathrm{Tr}(\rho_{AC}^2) \sim 1$.

Problem 5

Consider the following two-qubit quantum state:

$$|\psi\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle - |11\rangle\right)$$

$$(I\otimes Y)|\psi\rangle$$
$$= \frac{1}{2}\left[(I|0\rangle)\otimes(Y|0\rangle)+\cdots\right)$$
$$= \frac{1}{2}\left(i|01\rangle - i|00\rangle + i|10\rangle + i|11\rangle\right)$$

The **Pauli-Y gate** is defined as:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

or $I\otimes Y = \begin{pmatrix} Y & 0 \\ 0 & Y \end{pmatrix} = \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}$

$$(I\otimes Y)|\psi\rangle = \frac{1}{2}\begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}\begin{pmatrix} \\ \\ \\ \end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix} -i \\ \vdots \end{pmatrix} = \sim$$

Tasks:
Apply the operators $\mathbb{I}\otimes Y$ and $Y\otimes\mathbb{I}$ separately to the state $|\psi\rangle$, and compute the resulting two-qubit quantum states in the computational basis.

Problem 6

Consider a five-qubit quantum system initialized in the state:

$$|\psi\rangle = |+\rangle\otimes|0\rangle\otimes|1\rangle\otimes|0\rangle\otimes|1\rangle, \quad \text{where } |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$$

Tasks:
(1) A magnetic field acts only on the first qubit. The operation is described by the following unitary matrix:

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$

Compute the resulting state $|\psi_1\rangle$ of the five-qubit system after applying $U\otimes\mathbb{I}\otimes\mathbb{I}\otimes\mathbb{I}\otimes\mathbb{I}$ to the initial state $|\psi\rangle$.

$U|+\rangle = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ i & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1+i}{2}\begin{pmatrix} 1 \\ i \end{pmatrix}$ 归一 $|\psi_1\rangle = \frac{1+i}{2}(|00101\rangle + |10101\rangle)$

con 为 |0⟩, tar 不变
con 为 |1⟩, tar 翻转

(2) Now apply a controlled-NOT (CNOT) gate between the first and second qubits (first as control, second as target). Compute the new state $|\psi_2\rangle$ after this operation.

$$|\psi_2\rangle = \frac{1+i}{2}(|00101\rangle + |11101\rangle)$$

(3) Question: Is the final state **entangled**? If yes, identify which qubits are entangled.

$$|\psi_2\rangle = \underbrace{\frac{1+i}{2}(|00\rangle+|11\rangle)}_{\text{纠缠部分}}\otimes\underbrace{|1\rangle\otimes|0\rangle\otimes|1\rangle}_{\text{独立部分}}$$

由(2), 不可分离, 测量相关

Problem 7

$S_i = \frac{\hbar}{2}\sigma_i$ 一般以z为基矢

任何方向均是 $(|+\rangle_z \text{ or } |\uparrow\rangle)\frac{\hbar}{2}$ or $-\frac{\hbar}{2}(|-\rangle_z \text{ or } |\downarrow\rangle)$

An electron is initially prepared in an eigenstate of the spin S_z, with eigenvalue $\frac{\hbar}{2}$. Please determine the possible outcomes when measuring the spin component S_x, and calculate the corresponding probabilities.

$|\psi\rangle = |+\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

S_x本征态 $\begin{cases} |+\rangle_x = \frac{1}{\sqrt{2}}(|+\rangle_z + |-\rangle_z) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ |-\rangle_x = \frac{1}{\sqrt{2}}(|+\rangle_z - |-\rangle_z) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{cases}$

$\langle +_x|\psi\rangle = \left(\frac{1}{\sqrt{2}}(1 \quad 1)\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$

$P = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

$\langle -_x|\psi\rangle = \left(\frac{1}{\sqrt{2}}(1 \quad -1)\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}$

$P = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

$S_x|\lambda\rangle = \lambda|\lambda\rangle$. $\lambda = \pm\frac{\hbar}{2}$

e.g. $\frac{\hbar}{2}$下: $\frac{\hbar}{2}\sigma_x|+\rangle_x = \frac{\hbar}{2}|+\rangle_x \to \begin{pmatrix} 1 \\ 1 \end{pmatrix} \to \frac{1}{\sqrt{2}}(|+\rangle_z + |-\rangle_z)$

Problem 8

This spinor of an electron in the y-z plane of the Bloch sphere can be written as:

$$|\psi\rangle = \cos\left(\frac{\alpha}{2}\right)|0\rangle + i\sin\left(\frac{\alpha}{2}\right)|1\rangle\,, \quad \text{with } \alpha \in [0,\pi]$$

Please try to determine the possible outcomes when measuring the spin component S_y. Then, calculate the corresponding probabilities as functions of the parameter α.

$S_y = \pm\frac{\hbar}{2} \rightarrow S_y|\lambda\rangle = \lambda|\lambda\rangle$, $\lambda = \pm\frac{\hbar}{2}$

$|+\rangle_y = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, $|-\rangle_y = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

$P(S_y = +\frac{\hbar}{2}) = |\langle +_y|\psi\rangle|^2$, $P(S_y = -\frac{\hbar}{2}) = |\langle -_y|\psi\rangle|^2$

Problem 9

We consider a quantum entanglement swapping scenario involving four particles:

- Particle 1 is located in London,
- Particles 2 and 3 are in Shanghai,
- Particle 4 is in Sydney.

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$
$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

(or) 可反解出 $|00\rangle = \frac{|\Phi^+\rangle + |\Phi^-\rangle}{\sqrt{2}}$, $|11\rangle = \cdots$, $|10\rangle = \cdots$, $|01\rangle = \cdots$

Initially, the entangled states are:

- Particles 1 and 2 share the following entangled state:

$$|\psi\rangle_{12} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

- Particles 3 and 4 are in a non-standard entangled state:

$$|\chi\rangle_{34} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$

A Bell basis measurement is now performed on particles 2 and 3 in Shanghai.

Tasks:

1. Express the total initial state of the four particles $|\psi\rangle_{12} \otimes |\chi\rangle_{34}$ in terms of the Bell basis of particles 2 and 3.
2. After the Bell basis measurement on particles 2 and 3, determine the resulting state of particles 1 and 4 for each possible Bell outcome. 归一化
3. For each case, what local unitary operation should be performed on particle 1 (or particle 4) to convert the state of particles 1 and 4 into the maximally entangled Bell state $|\phi^+\rangle$? $\leftrightarrow |\Phi^+\rangle$

1. $|\psi_{12}\rangle \otimes |\chi\rangle_{34} = \frac{1}{2\sqrt{2}}(|0100\rangle + |0101\rangle + |0110\rangle - |0111\rangle - |1000\rangle - |1001\rangle - |1010\rangle + |1011\rangle)$

$= |\Psi^-_{23}\rangle|\Psi^{(1)}_{14}\rangle + |\Psi^+_{23}\rangle|\Psi^{(2)}_{14}\rangle + |\Phi^+_{23}\rangle|\Psi^{(3)}_{14}\rangle + |\Phi^-_{23}\rangle|\Psi^{(4)}_{14}\rangle$

2. $|\Psi^{(1)}_{14}\rangle = \frac{1}{2}(-|00\rangle - |01\rangle - |10\rangle + |11\rangle)$
$|\Psi^{(2)}_{14}\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle)$
$|\Psi^{(3)}_{14}\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)$
$|\Psi^{(4)}_{14}\rangle = \frac{1}{2}(-|00\rangle + |01\rangle - |10\rangle - |11\rangle)$

3. ① Hadamard门 $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ 对4.
② 调换"±"进行逆相位操作

$Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$, $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$

$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, $X = |1\rangle\langle 0| + |0\rangle\langle 1|$

灵活使用

$H|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$, $H|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$

☆ 即先进行H门操作，之后 $(Z\otimes X)|\Psi^-\rangle_{14}$
(若存在 $|01\rangle$, $|10\rangle$)

记得归一

Problem 10

We are given the following setup:

- Photons 2 and 3 (shared between Beijing and Shanghai) are in a partially entangled state with a hidden phase:

$$|\phi_{23}\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + e^{i\delta}|11\rangle\right)$$

- Photon 1 (in Beijing) is entangled with an ancilla qubit (photon 0), forming the state:

$$|\psi_{01}\rangle = \alpha|00\rangle + \beta|11\rangle$$

  where photon 0 is kept in a quantum memory in Beijing.

- A joint Bell basis measurement is performed on photons 1 and 2.

$$|\psi_{0123}\rangle = \frac{\alpha}{\sqrt{2}}|0000\rangle + \frac{e^{i\delta}\alpha}{\sqrt{2}}|0011\rangle + \frac{\beta}{\sqrt{2}}|1100\rangle + \frac{e^{i\delta}\beta}{\sqrt{2}}|1111\rangle$$
$$= \frac{1}{2}\left[|\Phi^+_{12}\rangle(\alpha|00\rangle_{03} + \beta e^{i\delta}|11\rangle_{03}) + |\Phi^-_{12}\rangle(\alpha|00\rangle_{03} - \beta e^{i\delta}|11\rangle_{03}) + \cdots\right]$$

Tasks:

1. Write down the total initial state $|\Psi\rangle_{0123} = |\psi_{01}\rangle \otimes |\phi_{23}\rangle$. Express it in terms of the Bell basis of photons 1 and 2.

$$|\psi_{03}^{11}\rangle = \alpha|00\rangle_{03} + \beta e^{i\delta}|11\rangle_{03} \quad \cdots$$

2. For each possible outcome of the Bell measurement on photons 1 and 2, determine the resulting joint state of photons 0 and 3.

3. Suppose Bob (in Shanghai) receives classical communication about the Bell measurement outcome. What local operation should he perform on photon 3 to restore the entangled state $|\psi_{03}\rangle = \alpha|00\rangle + \beta|11\rangle$ between photon 0 and photon 3?

恢复特定纠缠态

4. Discuss: if the phase δ is unknown, how does it affect the fidelity of the restored state? Can the entanglement still be used for reliable quantum teleportation or quantum key distribution?

Bob 的纠正操作

Alice 在 AB 系统上测量，得到结果：

- 若测到 $|\Phi^+\rangle_{AB}$，则 Bob 手中的粒子 C 自动变为 $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- 若测到 $|\Phi^-\rangle_{AB}$，则 Bob 拥有 $\sigma_z|\psi\rangle$
- 若测到 $|\Psi^+\rangle_{AB}$，则 Bob 拥有 $\sigma_x|\psi\rangle$
- 若测到 $|\Psi^-\rangle_{AB}$，则 Bob 拥有 $i\sigma_y|\psi\rangle$

Alice 将她的测量结果（2 个经典比特）通过经典信道发送给 Bob。

Bob 根据 Alice 传来的结果，对粒子 C 做如下幺正变换：

Alice	Bob
$\|\Phi^+\rangle_{AB}$	I
$\|\Phi^-\rangle_{AB}$	σ_z
$\|\Psi^+\rangle_{AB}$	σ_x
$\|\Psi^-\rangle_{AB}$	$\sigma_x\sigma_z = i\sigma_y$

完成后，Bob 手中粒子变为原始的 $|\psi\rangle$。

4. 对于 $|\Phi^+\rangle/|\Phi^-\rangle$: $\alpha|00\rangle_{03} + \beta e^{i\delta}|11\rangle_{03} = \phi_1$, $P = \frac{1}{2}$
对于 $|\Psi^+\rangle/|\Psi^-\rangle$: $\alpha e^{i\delta}|00\rangle_{03} + \beta|11\rangle_{03} = \phi_2$, $P = \frac{1}{2}$
保真度定义: $F = |\langle\psi|\phi_1\rangle|^2$

$$F_1 = |\langle\psi|\phi_1\rangle|^2 = \left||\alpha|^2 + |\beta|^2 e^{i\delta}\right|^2 = |\alpha|^4 + |\beta|^4 + \underbrace{|\alpha|^2|\beta|^2(e^{i\delta}+e^{-i\delta})}_{2|\alpha|^2|\beta|^2\cos\delta}$$

$F_2 = |\langle\psi|\phi_2\rangle|^2 = \cdots \Rightarrow F_1 = F_2$

$\Rightarrow F = |\alpha|^4 + |\beta|^4 + 2|\alpha|^2|\beta|^2\cos\delta$

$\delta = 0$, $F = 1$, 完美恢复

$\delta \neq 0$, $F < 1$, 不可靠

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos\theta - i\sin\theta$

$e^{i\theta\hat{\sigma}} = \cos\theta \cdot I + i\sin\theta \cdot \hat{\sigma}$